# ADRITA RAHMAN TORY

*Independent Researcher | AI Safety, Security & Alignment*
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## SUMMARY

Researcher at the intersection of cybersecurity and AI security, currently expanding my focus into AI Safety and Effective Altruism (EA). Over the past two years, I have co-authored 11 peer-reviewed papers and mentored 11 undergraduate researchers in frontier AI security. My recent work explores detecting covert collusion among AI agents. Dedicated to building safe, robust, and human-aligned intelligence systems with a long-term, impact-driven vision.

## RELEVENT PROJECTS

**Collusion Evidence Lab** — Apart Research sprint [GitHub] Oct 2026

- Tested simple ways to spot AI agents secretly coordinating prices, using only their logs. No single check worked everywhere, but combining them cut false alarms from 34% to 4%. It still misses cartels that never react to rivals.
- Built a free browser tool that shows the evidence to a human reviewer, so a person makes the final call.

**IDS/IPS Dataset Scoring Tool** — based on my Computers & Security paper [GitHub] 2026

- My research asked whether public intrusion-detection datasets cover the attacks that different industries really face. I then turned it into a free browser tool: pick an industry, and it scores 10 well-known datasets, or your own.

## RESEARCH EXPERIENCE

**Cyber Research Hub BUBT** (research lab) — Research Assistant and mentor Jul 2024 – Jul 2026
*Supervised by Dr. Khondokar Fida Hasan (UNSW)*

- Supported research on privacy-preserving 6G infrastructure, federated learning, and explainable and trustworthy AI in cybersecurity; ran experimental evaluations and assisted in building research prototypes.
- Ensured research activities complied with privacy, governance, and policy requirements across Australian and global contexts.
- **Mentored 11 undergraduate researchers on methods, experiments and writing. Three of our papers are already accepted.**
- Ran workshops for CSE students, including one on turning an observation into a research question.

**Bangladesh University of Business & Technology (BUBT)** — Research Assistant May 2025 – Aug 2026
*Supervised by Md. Saiful Rahman(BUBT)*

- Conducted focused research on AI-based cyber threat detection and adversarial AI, including comprehensive experimental evaluations
- Contributed to academic papers, preparing figures, summarising findings, and supporting writing and documentation
- Built research capacity among undergraduate students in cybersecurity at BUBT through workshops and one-on-one mentoring.

## EDUCATION

**Professional Master's in IT** — Jahangirnagar University Aug 2025 – 2026
*Thesis submitted, awaiting result: "A Multi-Signal Evidence Framework for Open-Set Network Intrusion Detection"*

**B.Sc. in Computer Science & Engineering** — BUBT 2021 – 2025
*CGPA 3.55/4.00. Capstone: an AI career-matching platform for students* [GitHub]

**Diploma in Computer Engineering** — Dhaka Mohila Polytechnic Institute 2016 – 2021

## PUBLICATIONS (11 PEER-REVIEWED-(Only Published and Accepted Papers are presented here))

### Journal Articles

**Tory, A. R.,** & Hasan, K. F. (2025). *An Evaluation Framework for Network IDS/IPS Datasets: Leveraging MITRE ATT&CK and Industry Relevance Metrics.* Computers & Security- Q1, **IF: 6.8. Published**

Hasan, K. F., Hughes, W., **Tory, A. R**., Campbell, C., & Turkay, S. (2025). *Game On: A Developmental Approach to Cyber Escape Room for Cybersecurity Governance and Policy Education.* Education Sciences, MDPI.**- Q1, IF: 3.5 Published**

### Conference Papers

**Tory, A. R.,** Hasan, K. F., Rahman, M. S., Koroniotis, N., & Moni, M. A. (2025). *Mind the Gap: Missing Cyber Threat Coverage in NIDS Datasets for the Energy Sector.* **Presented at:** *ICBDIML 2025*.

**Tory, A. R.,** Ali, A. S., Rahman, M. A., & Hasan, K. F. (2026). *Privacy, Robustness, and Fairness Trade-offs in Federated Intrusion Detection: Geometric Indistinguishability at the Aggregation Interface.* **Presented at:** *ML4CS 2026.*

Hasan, K. F., Hughes, W., & **Tory, A. R.** (2025). *Gamifying Cyber Governance: A Virtual Escape Room to Transform Cybersecurity Policy Education.* **Presented at***: ISSOTL 2025.*

Tonmoy, T. R., **Tory, A. R.,** Islam, M. R., Habib, M. A., & Rahaman, M. A. (2026). *Evaluating Provenance-Based Defenses Against Prompt Injection in LLM Agents: What Taint Tracking Can and Cannot See.* **Accepted at:** *IEEE CSDE 2026.*

Chakraborty, S., **Tory, A. R.,** Hasan, K. F., & Rahman, M. A. (2026). *An Empirical Evaluation of Quantum Kernel Stability in Network Intrusion Detection.* **Accepted at:** *IEEE iCOSTE 2026.*

Tonmoy, T. R., **Tory, A. R.,** & Hasan, K. F. (2026). *DevSecOps 2.0: Shift-Up and a Sec-Trust-Gov Model for AI-Enabled Software Development.* **Accepted at:** *IEEE ICCIT 2026.*

Rahi, S. M., Siddika, A., Maruf, I. **Tory, A. R**, H., Rahaman, M. A., Hasan, K. F., &. (2026). *A Decade of Healthcare Cyber Threats: Empirical Analysis, Evidence-Based Prioritisation, and AI Threat Model.* **Presented at:** *ML4CS 2026.*

Mukta, J. S., Sanjida, R., Tory, A. R., Rahman, M. S., & Hasan, K. F. (2026). *Preprocessing Failure and Adversarial Detection in Depthwise-Separable Edge Vision Systems.* **Presented at:** *ML4CS 2026.*

Khanom, S., Khan, S. A., **Tory, A. R.,** Hasan, M. A., & Hasan, K. F. (2026). *Multi-LLM Consensus Framework for Evaluating Banking-Sector NIDS Dataset Coverage of MITRE ATT&CK Techniques.* **Presented at:** *ML4CS 2026.*

## RELEVENT COURSES, BOOTCAMP, AWARDS & SERVICE

- **Technical AI Safety course**, BlueDot Impact (completed Oct 2026)
- **Career Bootcamp**, Centre for Effective Altruism (2026)
- Erasmus Mundus CyberMACS Scholarship (awarded 2026)
- Peer Reviewer, COMPAS 2025 — IEEE Computer Society Bangladesh Chapter
- Subreviewer / External Reviewer— STI 2024 and STI 2025
- Designed and facilitated research-training workshops for CSE students at BUBT, including “Cybersecurity Under the Umbrella: Paths, Reality, and Research Readiness” and “From Observation to Research Question”, covering research-gap identification and thesis/publication scoping

## CERTIFICATIONS

- CompTIA Security+ — CompTIA (2025)
- Cisco Network Security: Intrusion Detection and Prevention, By: Lisa Bock — LinkedIn Learning (2026)
- RAG, AI Apps, and AI Agents for Cybersecurity and Networking, By: Pearson and Omar Santos — LinkedIn Learning (2026)
- IT Essentials — Cisco (2020)
- National Certificate-IV in CBT&A Methodology for Trainers & Assessors — BTEB (2021)

## SKILLS

- **AI and ML:** PyTorch, TensorFlow, scikit-learn; adversarial robustness; federated learning with differential privacy; explainable AI; prompt injection and LLM-agent security
- **Security:** MITRE ATT&CK and ATLAS, OWASP Top 10 (incl. LLM), Wireshark, Zeek, Snort, Splunk; NIST AI RMF and EU AI Act
- **Research and code:** experiment design, statistics, literature review; Python, TypeScript/React, C++, SQL, Docker, Git, LaTeX

## LANGUAGES & ENGLISH PROFICIENCY

- **Bengali:** Native
- **IELTS (Academic):** Overall 7.0 | Reading 7.5 | Listening 8.0 | Speaking 6.5 | Writing 6.0

## PROFESSIONAL EXPERIENCE

**Software Engineer** *(Part Time)* | *Sun Co. Ltd., Dhaka, Bangladesh*
*Jan 2023 – Jul 2024*

**Intern, Software Engineering** | *Sun Co. Ltd., Dhaka, Bangladesh*
*Jul 2022 – Dec 2022*

**RPL/RTO Trainer** | *Bangladesh Technical Education Board, Dhaka, Bangladesh*
*Mar 2020 – Jun 2021*

**Guest Trainer, IT & Web Development** | *Ideal Institute of Technology, Rishda Bangladesh, Kaizen IT LTD, Technical Teachers' Training College, New Trizon IT — Dhaka, Bangladesh*
*2019 – 2022*

- Delivered Course wise training sessions in IT operations, web development, and graphic design across five institutes